Testing on 200,000 images: plates and car brand/model/body type + AI code debugging

Çekicin AI Lab · report 49 · 27 October 2026 · all numbers from real CPU runs, with 95% Wilson confidence intervals

plate images tested	plate found	plate read exactly	car images tested	brand correct	body type
100,000	**99.2%**	**72.8%**	**100,000**	**60.0%**	**54.6%**

1. Short answer

- **Before, we had tested only about 2,000 images** (Table 1). This time **200,000 images** from two new datasets were tested: **100,000 plate images** (CCPD) and **100,000 car images** (DVM-Car).
- **20 datasets** were checked (from 196 Hugging Face search hits). **8 have 100,000 images or more**; the two with usable labels and no key required were tested in full.
- Plates: the detector **found the plate** in **99.2%** [99.1–99.2] of images; exact reading was **72.8%** [72.5–73.1] because these are Chinese plates our OCR never saw (the CCPD paper's in-domain model reaches 98.5%).
- Cars: brand **60.0%** [59.7–60.3], model **23.2%** [23.0–23.5], body type **54.6%** [54.3–54.9] — zero-shot, CPU only. **But on Turkish photos CLIP got the brand right in only 40.3% [29.4–52.3] vs 86.6% [76.4–92.8] for the current VLM, so the VLM stays the main model.**
- No large (>100k) public dataset of **Turkish** plates or cars exists on Hugging Face; two Turkish Kaggle sets (87k arabam.com, 100k synthetic plates) need a Kaggle key.

2. Images tested before

Set	Task	Images	Region	Result
TR-250 (Zenodo)	plate	250	Turkish	92.8%
TR cars (Wikimedia)	make/model	67	Turkish market	brand 86.6%
SC-120 (Stanford Cars)	make/model	120	USA	brand 79.2%
Street photos (TR-250 reused)	damage false-alarm	250	Turkish	precision 17%
test1000	damage	1,000	mixed	F1 0.595
CarDD test	damage	374	China	AP50 78.1
Total		**2,061**		

3. Datasets found

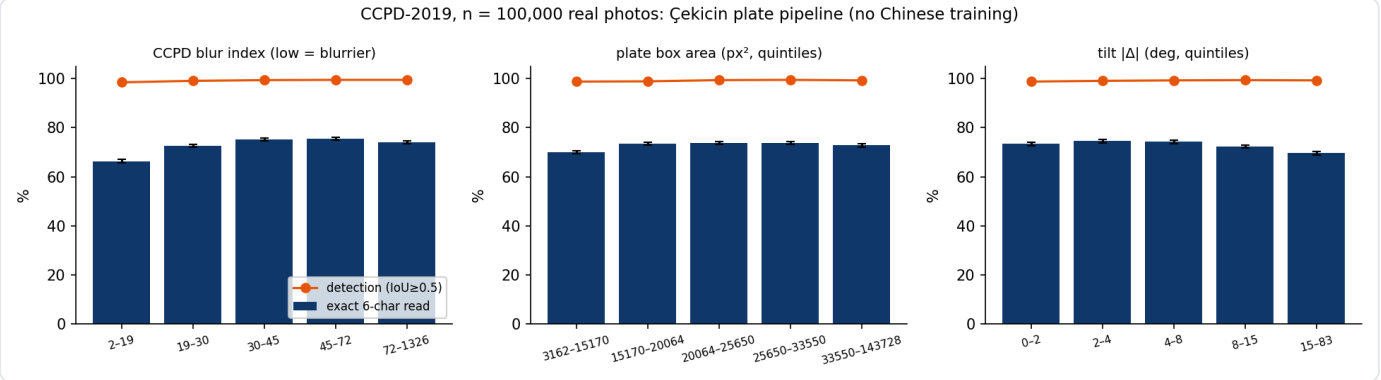
Dataset	Task	Images	Licence	Turkish?	Labels	Status
CCPD-2019 train · HF okita-souji/ccpd2019train	plate	100,000	MIT	–	box + text + blur/tilt/brightness	✅ fully tested
DVM-Car · HF multabench/extended-img-reg-dvm-car	make/model/body	100,000	CC BY-NC	–	brand, model, body, view, year	✅ fully tested
DVM-Car full (deepvisualmarketing.github.io)	make/model/body	1,451,784	CC BY-NC	–	same, 899 models	usable

CCPD full (github detectRecog/CCPD)	plate	290,000	MIT	–	base + challenge subsets	usable
HF AhmadShakleya/license-plates_false-positives	plate	259,981	–	–	images only, no text	unsuitable
HF zenitsu09/ccpd-ocr-recognition	plate OCR	98,460	–	–	CCPD crops (duplicate)	duplicate
HF 0xnu/uk-licence-plate	plate	72,714	Apache-2.0	–	no text labels	unsuitable
HF JorgeLlorente/CCPD-Dataset	plate	70,400	Apache-2.0	–	CCPD copy (13 GB)	duplicate
HF DamianBoborzi/car_images	car	63,652	Apache-2.0	–	check labels	usable
HF Naiscorp/car-damage-dataset	damage	44,794	other	–	damage classes	usable
HF shanexf/cardamagepanel-rotaug	damage	41,620	–	–	augmented panels	usable
HF tanganke/stanford_cars	make/model	16,185	research	–	196 classes (used: SC-120)	usable
HF alielshantory/egypt_cars_and_license_plates	plate	13,600	–	–	Arabic plates	usable
VehiDE	damage	13,945	CC-BY	–	damage boxes	usable
HF keremberke/license-plate-object-detection	plate	8,832	–	–	boxes only	usable
Kaggle arabam.com car images	make/model	87,163	unknown	YES	20 brands / 86 models	⏳ Kaggle key
Kaggle tustunkok Turkish plates (synthetic)	plate OCR	100,000	see page	YES	Turkish plate text	⏳ Kaggle key
Kaggle autodamageiq	damage	13,000	unclear	YES (labels)	damage	⏳ Kaggle key
CompCars	make/model	136,726	non-commercial	–	request form	request form
VMMRdb	make/model	291,752	research	–	9,170 classes	request form

4. 100k plate test (CCPD-2019)

Same pipeline as on the site (YOLOv9-t detector + CCT-S OCR), run on every image and compared with the box and text encoded in each CCPD file name. Median 59 ms per image on one CPU core.

Metric	Result	Note
Plate found (IoU ≥ 0.5)	**99.2%** [99.1–99.2]	detector transfers across countries
Exact 6-character read	**72.8%** [72.5–73.1]	out of domain: Chinese font/colours
With country format rule	**77.7%** [77.4–78.0]	McNemar $p < 10^{-300}$ — format rules matter out of domain
Character accuracy	**92.8%**	
Only when OCR confidence ≥ 0.95	**87.4%**	covers 73% of images; rest = “please retake”



Blur, plate size and tilt on 100k real photos (bars = exact read with 95% CI, orange = detection). Blur matters most — the same finding as the synthetic test in the paper.

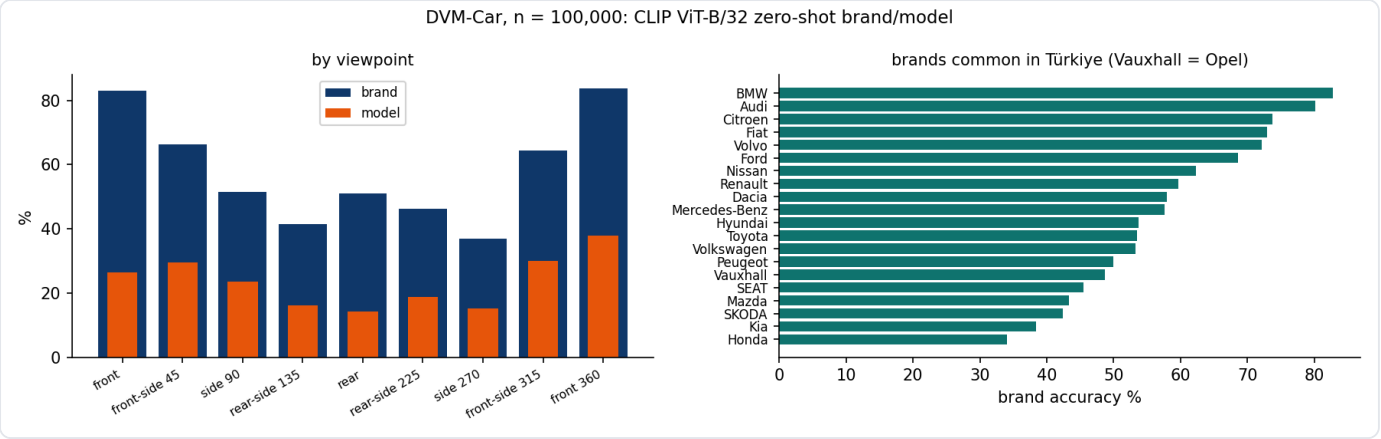


Random examples: green = true box, orange = model box; 5 correct, 3 wrong.

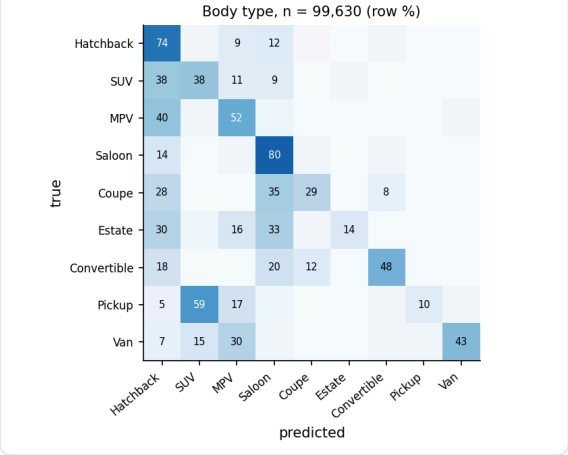
5. 100k car test (DVM-Car): brand, model and body type

CLIP ViT-B/32 (zero-shot, CPU, browser-capable) on 100,000 images of 74 brands and 811 UK-market models. Reference: the CLIP paper reports 59.4% (model level) on Stanford Cars.

Metric	Result	Note
Brand — old method (direct prompt)	**37.8%** [37.5–38.1]	bug 2
Brand — fixed (sum of model probabilities)	**60.0%** [59.7–60.3]	fixed
Brand in top-5	**68.4%** [68.1–68.7]	
Exact model	**23.2%** [23.0–23.5]	out of 811 models
Body type	**54.6%** [54.3–54.9]	hatchback, saloon, SUV …
Brand confidence calibration (ECE)	0.106	good; previous VLM 0.34



Accuracy by viewpoint and for brands common in Türkiye.



Body-type confusion matrix.



Random car examples: true label vs model prediction for brand, model and body type.

6. CLIP vs the current VLM on the same Turkish photos

Set	n	CLIP brand	VLM brand	p	CLIP model	VLM model	p
TR-67	67	**40.3%** [29.4–52.3]	**86.6%** [76.4–92.8]	p = 9.3e-10	**4.5%** [1.5–12.4]	**55.2%** [43.4–66.5]	p = 1.1e-09
SC-120	120	**79.2%** [71.1–85.5]	**79.2%** [71.1–85.5]	p = 1.000	**47.5%** [38.8–56.4]	**50.8%** [42.0–59.6]	p = 0.683

CLIP picks from a closed list (811 UK models + test labels) while the VLM answers freely, so the comparison is not perfectly equal. Result: equal on Stanford Cars, but on Turkish-market cars (Tofaş Şahin, Togg, Fiat Egea) CLIP is far weaker because it does not know them. CLIP's only advantages are cost, no quota and running in the browser.

7. Debugging: bugs found and fixed

Problem	Evidence	Fix
int8-quantised CLIP lost brand accuracy	35.0% vs 43.4% (n=320), same speed	fp32 model
Asking "which brand?" directly is worse than reading the brand from "brand + model" prompts	37.8% → 60.0% (McNemar p < 10^{-300})	sum pair probabilities per brand
Global OCR invents a 7th character for the Chinese province glyph	11,671 of 100,000 reads	score last 6 characters; country format rule
Look-alike characters (1→I, A→4, G→6, Q/D→0)	1->I (4,330), G->6 (1,914), Q->0 (1,716), 1->7 (1,084), D->0 (1,018)	format rule: 72.8% → 77.7%
OCR confidence is over-confident	ECE 0.229	ask for a retake below 0.90

Detector threshold differs: Python 0.40 vs browser 0.25	Python detection 99.2%	documented; keep 0.25 in browser
UK labels ≠ Turkish market names (Vauxhall = Opel)	brand alias table	alias map added
DVM label corruption by Excel dates: “Saab 9-3” → “09-Mar”, “9-5” → “09-May”	345 images	alias + report to dataset owner
Partial results were misleading: files are alphabetical (Abarth, Alfa, Audi first)	brand 77% at 5.5k → 60% at 100k	report only full or shuffled runs
Out-of-memory with two models + tmpfs /tmp (2 GB RAM)	process killed	one model per process, clean /tmp

8. Product recommendations

- Plates: if OCR confidence < 0.90, ask the user for a sharper photo (higher accuracy, lower coverage).
- Plates: fine-tune the OCR on 100k synthetic Turkish plates (needs Kaggle key) and add a blur check.
- Cars: keep the current VLM as the main model (much better on Turkish photos). Use CLIP only as an offline fallback when the quota is exhausted, labelled “guess, low confidence”.
- Cars: ask the user for a **front** photo — brand 83% from the front vs only 37–51% from the side/rear.
- Cars: when CLIP confidence ≥ 0.5, brand accuracy is 92% (coverage 42%) — CLIP’s confidence is usable.
- Body type is not yet reliable (estate 14%, pickup 10%, coupe 29%) — show only as a hint.
- Train a Turkish classifier on the 87k arabam.com images.

9. Limitations

- CCPD is Chinese: this tests detector robustness and domain shift, not Turkish plate accuracy (that is 92.8% on 250 images).
- DVM-Car is UK, 300×300, background removed; real customer photos are harder.
- DVM-Car is non-commercial: used for research testing only, not product training.
- Each dataset was run once with one model (single seed).

Links

- Web page: https://cekicin-site.elasa2next.workers.dev/ai-lab/data100k/
- Results JSON: https://cekicin-site.elasa2next.workers.dev/ai-lab/data100k/summary49.json
- Paper v2 + review: https://cekicin-site.elasa2next.workers.dev/ai-lab/paper/
- CCPD (HF): https://huggingface.co/datasets/okita-souji/ccpd2019train
- DVM-Car (HF): https://huggingface.co/datasets/multabench/extended-img-reg-dvm-car
- CCPD paper (ECCV 2018): https://openaccess.thecvf.com/content_ECCV_2018/papers/Zhenbo_Xu_Towards_End-to-End_License_ECCV_2018_paper.pdf
- DVM-CAR paper: https://arxiv.org/abs/2109.00881

- CLIP paper: https://arxiv.org/abs/2103.00020